



Cyber Security & Digital Forensics

CSE - 4207

Lecture - 06

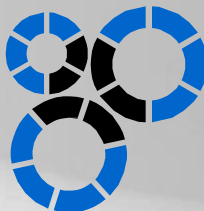
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Outline

- Introduction to Communication Security
- RSA
- Encryption
- Decryption
- Diffie and Hellman Key Exchange
- Bucket Brigade Attack



Introduction to Communication Security



A) Secret key (symmetric) cryptography. SKC uses a single key for both encryption and decryption.



B) Public key (asymmetric) cryptography. PKC uses two keys, one for encryption and the other for decryption.



C) Hash function (one-way cryptography). Hash functions have no key since the plaintext is not recoverable from the ciphertext.

RSA

- First practicable publickey cryptosystems
- Encryption key-**public**
- Decryption key-**private**
- Invented by Ron Rivest, Adi Shamir and Leonard Adleman
- RSA algorithm has 3 steps:
 1. Generation of public and private keys
 2. Encryption
 3. Decryption



RSA - Generation of Public & Private Keys

Pick two prime numbers p & q



$p = 3$ and $q = 11$

Calculate $n = p * q$



$n = 33$

Calculate $\phi(n) = (p-1)*(q-1)$



$\Phi(n) = 20$

Choose integer e such that
 $1 < e < \phi(n)$ and $\text{GCD}(e, \phi(n)) = 1$



$e = 7, 11, 13, 17, 19$

$e = 7$

RSA - Generation of Public & Private Keyes

$$e * d \pmod{\varphi(n)} = 1$$



$$7 * d \pmod{20} = 1$$

Public Key = {e, n}
Private Key = {d, n}



$$(7 * d) / 20 = 1$$



$$d = 3$$

RSA- Encryption

$$C = M^e \bmod n$$

Where,

C = Cipher Text

M= Original Message

Public Key = {e, n}

Let, M = 31 and Public Key= {7, 33}

$$C = 31^7 \bmod 33 = 4$$

RSA- Decryption

$$M = C^d \bmod n$$

Let, Cipher Text=4, Private Key= {4, 33}

$$M = 4^4 \bmod 33 = 31$$



Why RSA is Secure?

- Factoring an integer with at least 512-bit is very hard!
- But if you can factor big number n then given public key $\langle e, n \rangle$, you can find d , and hence the private key by:
 - Knowing factors p, q , such that, $n = p \times q$
 - Then compute $\phi(n) = (p-1)(q-1)$
 - Then find d such that $e \times d = 1 \bmod \phi(n)$

Diffie and Hellman Key Exchange

- First published public-key algorithm
- By Diffie and Hellman in 1976 along with the exposition of public key concepts
- Used in a number of commercial products
- Practical method to exchange a secret key securely that can then be used for subsequent encryption of messages
- Security relies on difficulty of computing discrete logarithms

Diffie and Hellman Key Exchange

Publicly known numbers

q = Prime number, of at least 300 digits

α = an integer that is a primitive root of q , often a small number

User C

Knows q, α, Y_A, Y_B

Must calculate

$$X_B = \text{dlog}_{\alpha, q}(Y_B)$$

User A

Selects a number $X_A < q$

Now has Y_B sent by User B

$$s = Y_B^{X_A} \bmod q$$

$$Y_A = \alpha^{X_A} \bmod q$$

$$Y_B = \alpha^{X_B} \bmod q$$

User B

Selects a number $X_B < q$

Now has Y_A sent by User A

$$s = Y_A^{X_B} \bmod q$$

Diffie-Hellman Example

Have

- Prime number $q = 353$
- Prime number $\alpha = 3$

A and B each compute their public keys

- A computes $Y_A = 3^{97} \bmod 353 = 40$
- B computes $Y_B = 3^{233} \bmod 353 = 248$

Then exchange and compute secret key:

- For A: $K = (Y_B)^{X_A} \bmod 353 = 248^{97} \bmod 353 = 160$
- For B: $K = (Y_A)^{X_B} \bmod 353 = 40^{233} \bmod 353 = 160$

Attacker must solve:

- $3^x \bmod 353 = 40$ which is hard
- Desired answer is 97, then compute key as B does



Diffie-Hellman Quiz

Alice and Bob agree to use
prime $q = 23$ and primitive root $\alpha = 5$

Alice chooses secret $a = 6$

Bob chooses secret $b = 15$

What number does Alice send Bob?

What number does Bob send Alice?

Diffie-Hellman Security

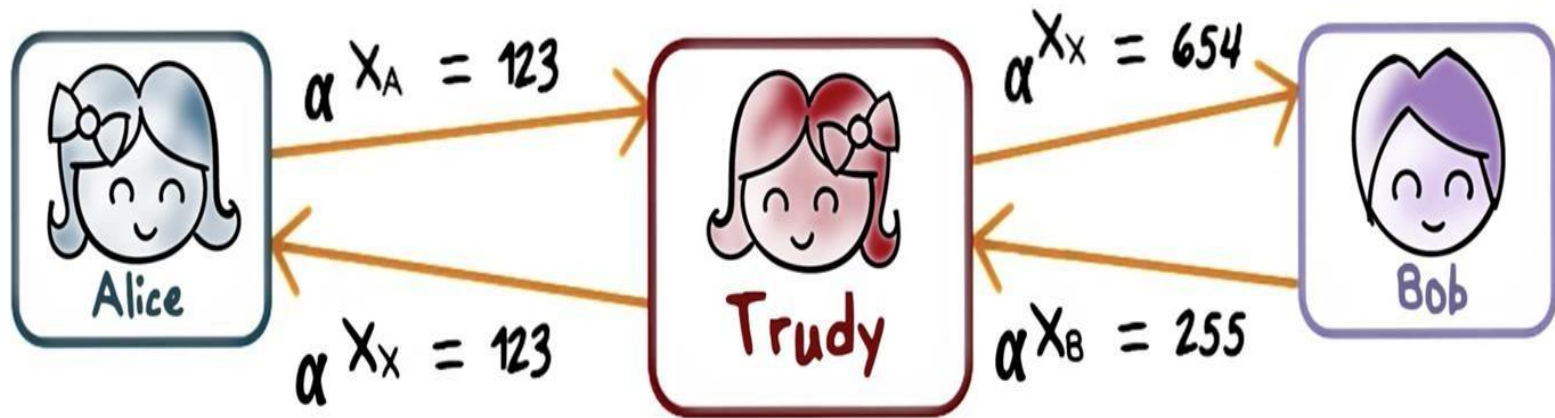
- Shared key (the secret) itself never transmitted
- Discrete logarithm is very hard
- $Y = \alpha^X \bmod q$
- Conjecture: given Y , α , and q , it is extremely hard to compute the value of X because q is a very large prime (discrete logarithm)

⦿ Diffie-Hellman Limitations

- Expensive exponential operation
 - DoS possible
- The scheme itself cannot be used to encrypt anything - it is for secret key establishment
- No authentication, so you cannot sign anything



Bucket Brigade Attack, Man-in-the-Middle(MIM)



$$\begin{matrix} X_A & X_X & X_X & X_B \\ 654^{X_A} = 123^{X_X} & 255^{X_X} = 654^{X_B} \end{matrix}$$

Trudy plays Bob to Alice and Alice to Bob



Thank You

